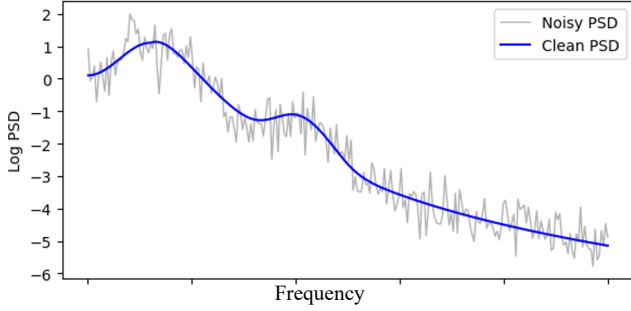


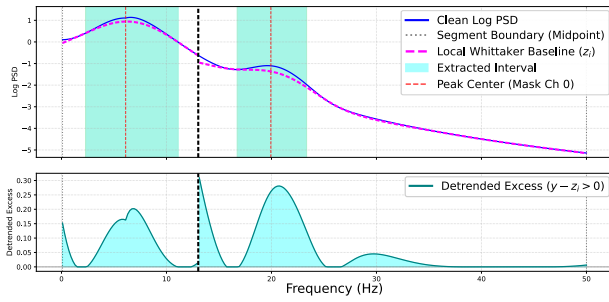
Automatic segmentation of Bumps in neural data PSD

A. Generation of synthetic data for supervised learning

a) Example of generated noisy PSD



b) Segmentation of bumps frequency intervals



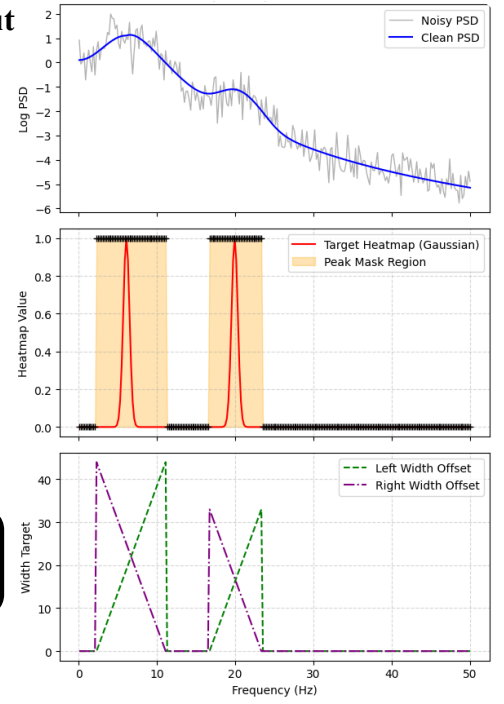
c) Neural Network input

Input: Noisy PSD

Supervised data:
clean PSD

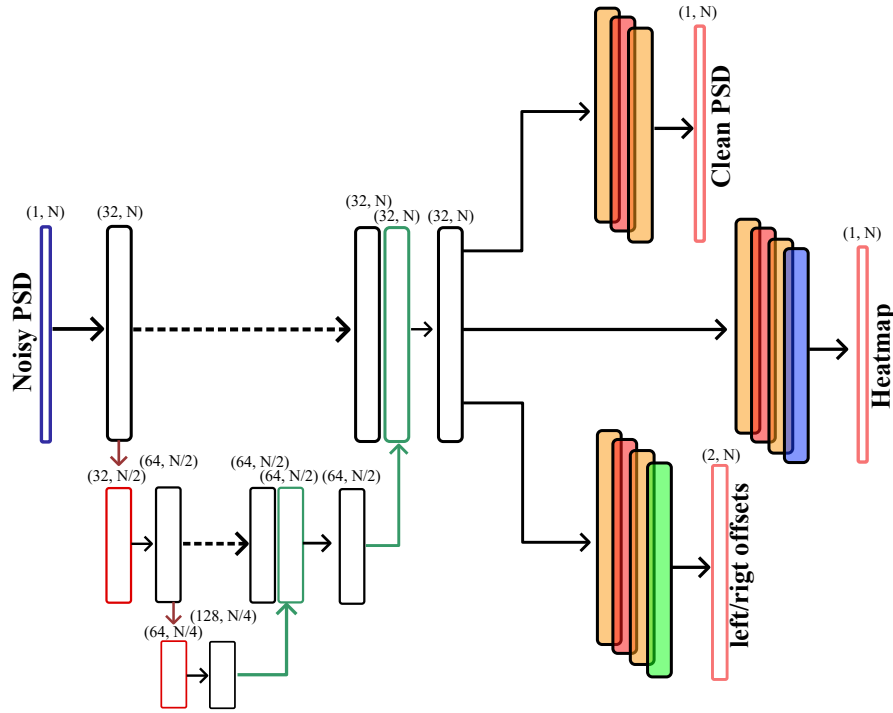
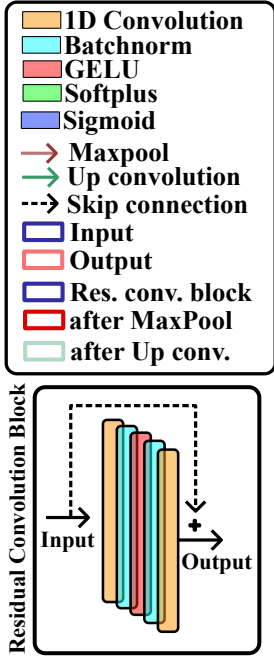
Supervised data:
Target heatmap

Supervised data:
left and right
widths offset



B. Neural Network architecture and Multitask loss

a) Model Architecture



b) Losses

- PSD MAE
- PSD derivative MAE
- PSD second derivative MAE

Center Focal Loss

- offsets MAE
- background

C. Results

a) Model Evaluation on test set

Loss	Value
PSD MAE	
PSD derivative MAE	
PSD 2nd derivative MAE	
Center Focal Loss	
Left Offset MAE	
Right Offset MAE	
Background	

b) Example of model inference

